# A Quick Java Swing Tutorial

# Introduction

- Swing – A set of GUI classes
  - Part of the Java's standard library
  - Much better than the previous library: AWT
    - Abstract Window Toolkit

- Highlights
  - A rich set of widgets
    - Widget: Any GUI element (also called: components)
  - Contents and shape are separated (MVC support)o
  - Fine-grained control over the behavior and  look and feel
  - Platform independent
    - Isolates the programmer from the operating system's GUI

# Swing Components

- Containers
  - Contain and manage other components.
  - Top Level/Internal
  - Examples: `JFrame` (Top Level), `JScrollPane`, `JPanel`.

- Basic controls
  - Atomic components
  - Used for showing ouput and/or getting some input
  - Inherits JComponent
  - Examples: `JButton, JLabel, JTextArea, JTable, Jlist`

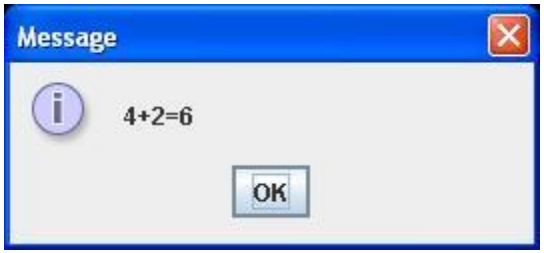
- Usually every Swing class extends the corresponding AWT class
  - For backward-compatibility reasons

# My First Swing Program

```
import javax.swing.*;
import java.awt.BorderLayout;

public class First {
  public static void main(String[] args) {
    JFrame frame = new JFrame("My First Frame");

    // operation to do when the window is closed.
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    frame.getContentPane().setLayout(new BorderLayout());
    frame.getContentPane().add(new JLabel("I Love Swing"),
       BorderLayout.CENTER);
    frame.pack();
    frame.setVisible(true);
  }
}
```

# Top Level Containers: JDialog

- **`javax.swing.JDialog`**:
  - More simple and limited than frames
  - Typically used for showing a short message on the screen
  - Also has a border and a title bar
  - May have an owner
    - If the owner is invisible the dialog will also be invisible
  - Use the static method of JoptionPane to show standard dialog boxes:
    `JOptionPane.showMessageDialog(null, "4+2=6");`



# Top Level Containers: JFileChooser

- **`javax.swing.JFileChooser`**:
  - Allows the the user to choose a file
  - Supports “open” and “save”: `showOpenDialog()`,`showSaveDialog()`h

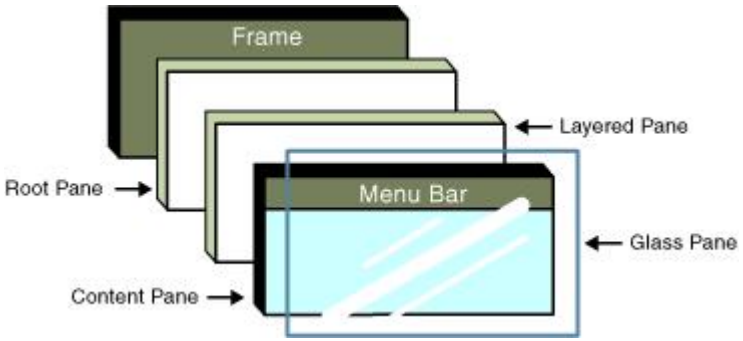
```
JFileChooser fc = new JFileChooser();
int returnVal = fc.showOpenDialog(null);
if(returnVal == JFileChooser.APPROVE_OPTION)
   System.out.println("File: " + fc.getSelectedFile());
```

# Top Level Containers: JFrame

- **`javax.swing.JFrame`**:
  - Top-level window with a title and a border.
  - Usually used as a program's main window

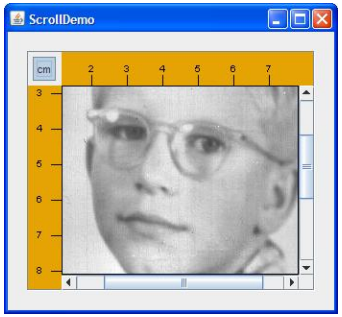
# More on JFrame

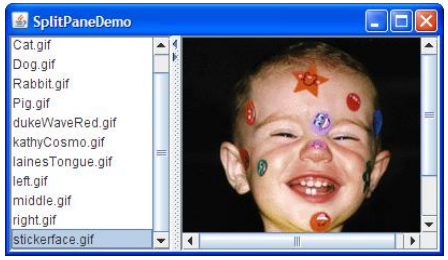
- Made of several layers
- Widgets are added to the Content Pane layer.
  - Use `getContentPane()` to obtain it
- Other layers are used for customizing the window's appearence

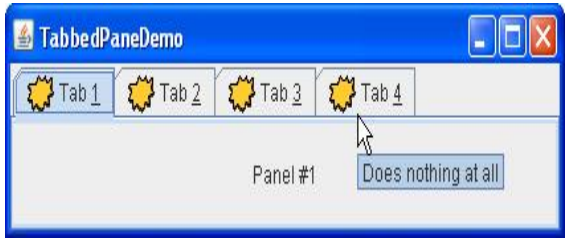


# Internal Containers

- Not Top level containers
- Can contain other non-top level components
- Examples:
  - `JScrollPane`: Provides a scrollable view of its components
  - `JSplitPane`: Separates two components
  - `JTabbedPane`: User chooses which component to see



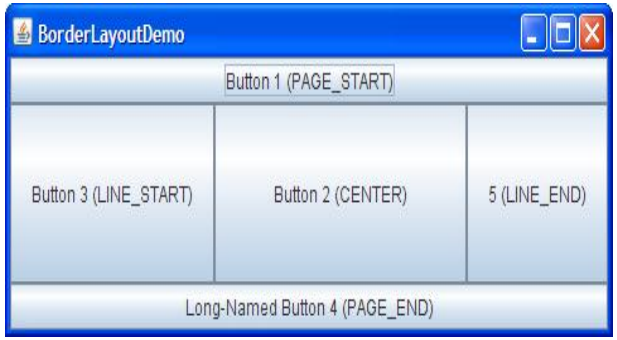
# Containers - Layout

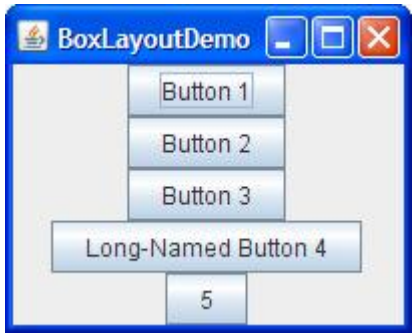
- Each container has a layout manager
  - Determines the size, location of contained widgets.

- Setting the current layout of a container:
  *`void setLayout(LayoutManager lm)`*

- *`LayoutManager`* implementing classes:
  - `BorderLayout`
  - `BoxLayout`
  - `FlowLayout`
  - `GridLayout`

# Containers - Layout



# Swing Components

**Basic Controls**

Simple components that are used primarily to get input from the user; they may also show simple state.

112kB

Middle button

JButton

Chin
Glasses
Hair
Teeth

JCheckBox

Pig
Bird
Cat
Dog
Rabbit
Pig

JComboBox

Martha Washington
Abigail Adams
Martha Randolph
Dolley Madison
Elizabeth Monroe
Louisa Adams

JList

A Menu | Another Menu
A text-only menu item Alt-1
Both text and icon
A radio button menu item
Another one
A check box menu item
Another one
A submenu

JMenu

Bird
Cat
Dog
Rabbit
Pig

JRadioButton

Frames Per Second
0 10 20 30

JSlider

Date: 07/2006

JSpinner

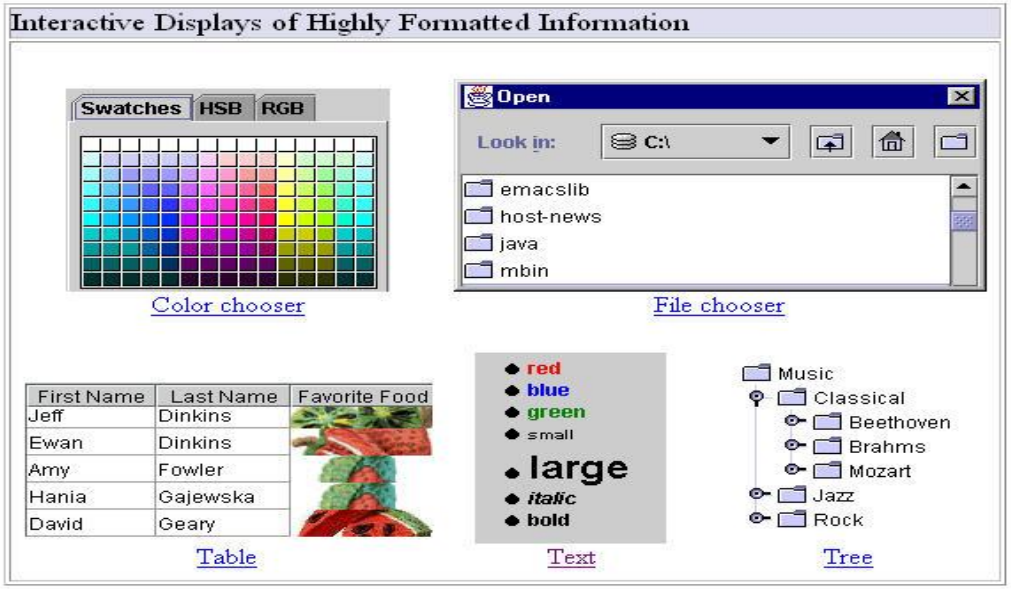
City: Santa Rosa

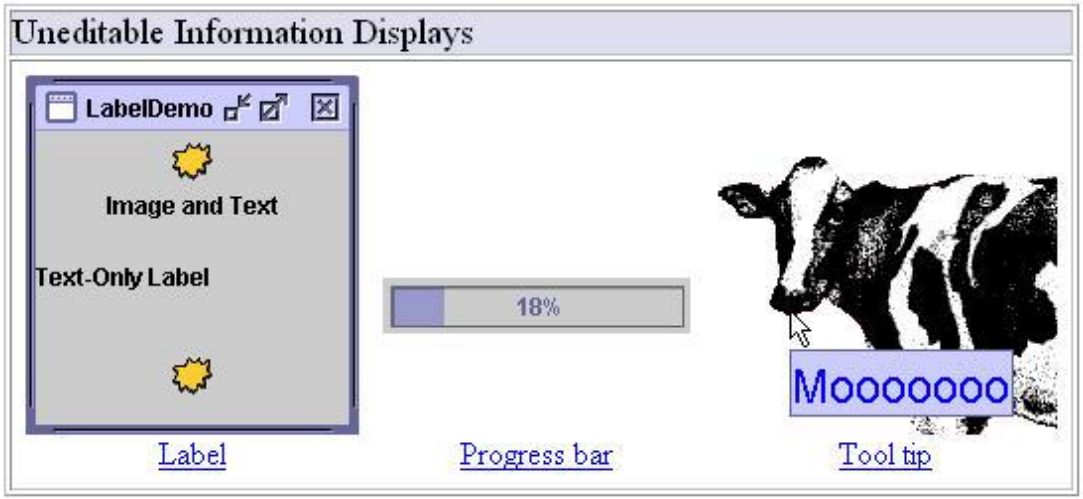
JTextField

Enter the password: •••••••

JPasswordField

# Swing Components



# First Swing Program Revisited

```
import javax.swing.*;
import java.awt.BorderLayout;

public class First {
  public static void main(String[] args) {
    JFrame frame = new JFrame("My First Fra

    // operation to do when the window is cl
    frame.setDefaultCloseOperation(JFrame  ER_ON_CLOSE);

    frame.getContentPane().setLayout(new BorderLayout());
    frame.getContentPane().add(new JLabel("I Love Swing"),
       BorderLayout.CENTER);
    frame.pack();
    frame.s  Visible(true);
  }
}
```

Create a frame

Choose the border layout

Create a text label

Add the label to the content pane

Specify CENTER as the layout position

# Input

- So we now know how to present widgets on the screen
- A program also needs to react to the user's actions
- Examples:
  - When the user presses a button we want to save a file
  - When the user closes the program we want to ask “are you sure?”
  - ...
- Swing mechanism: Events and Listeners

# Events, Listeners

- Swing defines all sorts of Listener interfaces
  - E.g.: `ActionListener, MouseMotionListener, ***WindowListener,*** ...`

```
public interface ActionListener extends EventListener {
  public void actionPerformed(ActionEvent e);
}

public interface MouseMotionListener extends EventListener {
  public void mouseDragged(MouseEvent e);
  public void mouseMoved(MouseEvent e);
}
```

- There are default (empty) implementations for many of the listeners
  - E.g.: `MouseMotionAdapter, ***WindowAdapter***`

# Events, Listeners (cont.)v

- A listener is an object that implements a listener interface
- If we need to react to an event (on a certain widget) we register a listener object with that widget
- E.g.: `addActionListener()` registers an action listener with its receiver:

```
JButton button = new JButton();
ActionListener listener = ...;
button.addActionListener(listener);
```

- When an event occurs, all registered listeners are notified
  - The appropriate listener method (e.g: `actionPerformed()`) is invoked
  - An object describing the event is passed as a parameter

# Event Handling Demo: GUI

# Event Handling Demo: Code

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Events implements ActionListener  {
   public Events() {
    JFrame frame = new JFrame("Events");
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    frame.getContentPane().setLayout(new FlowLayout());
    JButton b = new JButton("Click me!");
    b.addActionListener(this);
    frame.getContentPane().add(b);

    frame.pack();
    frame.setVisible(true);
  }
  public void actionPerformed(ActionEvent e) {
    JOptionPane.showMessageDialog(null, "Thank you");
  }
  public static void main(String[] args)  { new Events(); }
}
```

# Inner Classes

- Nested within another classes
- Instance specific:
  - Has access to methods & fields of the object that created it
  - => An inner class has TWO this variables
- Can be static
  - Can access only static members and methods only
  - A static method cannot create a non-static inner class

# Local Classes

- Same as inner classes but defined inside a method
- Has access to local variables of the enclosing method
  - Only if the variable is defined as final
- Can be anonymous
  - Doesn’t have a name.

# Event Handling Demo: Local Class

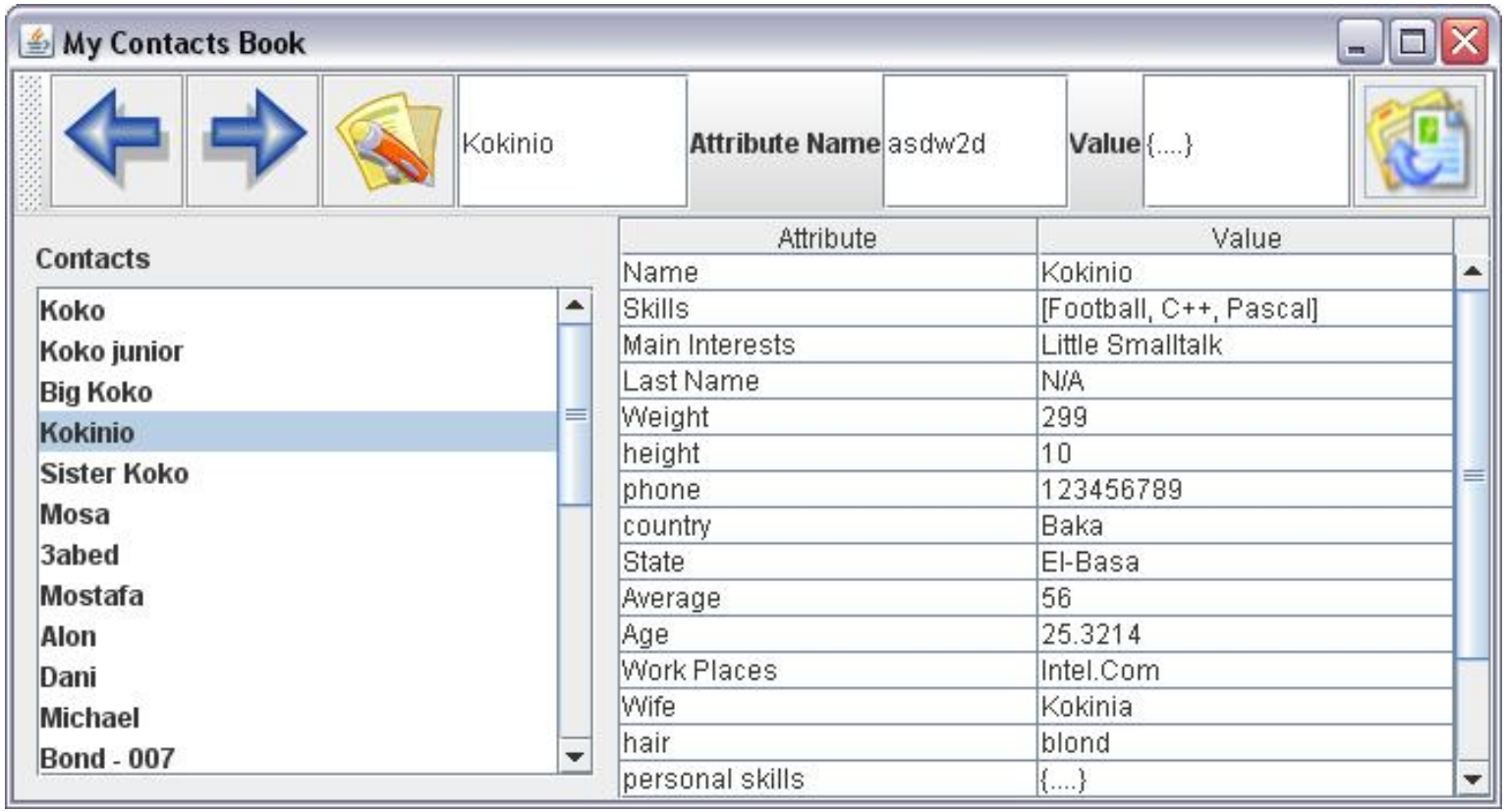
```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Events {
   public Events() {
     JFrame frame = new JFrame("Events");
     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

     frame.getContentPane().setLayout(new FlowLayout());
     JButton b = new JButton("Click me!");
     b.addActionListener(new ActionListener() {
       public void actionPerformed(ActionEvent e) {
         JOptionPane.showMessageDialog(null, "Thank you");
       }
     });
     frame.getContentPane().add(b);

     frame.pack();
     frame.setVisible(true);
   }
}
public class EventsTest {
   public static void main(String[] args)  { new Events(); }
}
```

# Appendix:
# Contact Book Program

# Swing's JTable

- Each `JTable` has a `TableModel` object that holds the data shown by the table.
- `DefaultTableModel` is a default implementation of `TableModel`
  - By default it makes all the cells editable
- We can customize its behavior by subclassing
  - Or we can implement `TableModel` from scratch
- *`TableModelListener`* - a listener interface
  - Notified of changes in the model
- Use `TableModel.addTableModelListener()` to register a listener object

# Contacts Book example overview

- Each contact has a name and a list of attributes
  - Each attribute has a name and a value.
- Operations:
  - Add contact
    - Contact name is immutable
  - Add attributes
    - Attribute is identified by it's name, the attribute name is immutable, it's value can be changed
- Classes:
  - `Contact`: represents a contact and maintains list of attributes
  - `Contact.Attribute`: Inner class that represents contact attribute.
  - `ContactTableModel`: model class for the table data
    - Doesn't allow to modify attribute names
  - `ContactBook`: represents the contact book widget.

# Contacts Book example overview